R16.4a Schumacher 2023 WRF Reproduction and Comparison: A Stage Report Paused Pending New Claude and GPT Models

PETW WRF research line, internal stage report

October 20, 2026 (Eastern time)

Abstract

This stage was paused, not closed, on October 20, 2026. PI decided to return after the next generations of Claude and GPT models are released. The exp-A scenario approximates the authors' early regional statistics and aspects of the gravity-wave-breaking downdraft, but its cumulative strong-wind footprint and first pulse remain weaker than the authors' simulation. A frozen operational rule labels the tested members "systematically weaker"; it does not identify the cause. Double-precision runs crashed during nested initialization, so precision has not been tested. All same-UTC-time comparisons carry `phase_match_unavailable`. This internal report preserves the evidence, adjudications, failures, and restart entry. `scientific_claim_approved=false`; none of these comparisons is an approved design result.

1 Question and evidence boundary

Schumacher, Childs, and Adams-Selin[2] simulated the June 9, 2020 Akron downburst with WRF 4.0.3 (configuration: PDF p. 4, Table 1). Exp-A used that version, the authors' Dryad initial conditions, a C0 boundary, and their namelist. Its formal Chinese name translates as "statistically equivalent author-equivalent scenario", but this is only a scenario label: no equivalence test or frame-by-frame reproduction was performed. The former R16.4b perturbation study now falls under R16.4a1, without relocating its files. The Dryad nwyu, swph, cph, and sedg series are *model tslist points*, not observed stations (paper Fig. 8: PDF p. 8, journal p. 782). The paper's simulated track lies about 60 km northeast of the observed region (PDF p. 7, journal p. 781); this is distinct from the distance metric used below. Same-UTC-time maps do not establish phase alignment: `phase_match_unavailable`. Time labels 05:15, 05:45, 05:59, 06:10, 06:30 and 06:45 below are the frozen metric or source-figure UTC identifiers; local Eastern times are respectively 01:15, 01:45, 01:59, 02:10, 02:30 and 02:45. Figure axes retain their source timestamps.

2 Configuration, operators, and preregistered tests

Exp-A was integrated in three segments (A, B, C). The configuration audit examined namelist, eta levels, GFS product and cycle, and real version. Matching early domain statistics is not identity of all storm-scale variables. For the frozen comparison, A is the cumulative event-region area where the minute-resolved, 10-m time-step maximum wind reaches 35 m/s during 05:15–06:30 model UTC. B is the maximum event-region wind in the first pulse (05:15–05:59). The authors' point value supplies a lower bound on their regional B, not an exactly matched regional measurement. The third criterion is the instantaneous lowest-layer area at 05:45 where wind reaches 35 m/s; it is missing for P2 after raw outputs were removed. Cumulative footprint and instantaneous area cannot be interchanged. The frozen coverage gates were A $\geq$ 403 km^2, B $\geq$ 44.4 m/s, or lowest-layer area $\geq$ 5 km^2; the six-member weaker gate required A $\leq$ 300 km^2 and B $\leq$ 42.4 m/s. These are operational gates, not a statistical significance test. P2 changed MPI layout and later switched platforms, confounding

a single-factor interpretation. E2c moved to CRC without added perturbation; E2p1–p3 added seed-specific 0.01 K temperature perturbations. E1 changed both the boundary version generated by WRF 4.0.3 and the cold-start chain. The low-level readout method was registered before mature E-member data appeared: it compares 150-m maximum wind and the increase from 500 to 150 m at 05:45 and 06:10. Its range contains exp-A and four E2 members. A double-precision build changed RWORDSIZE and converted inputs to double; the cold start did not complete nested initialization. Root-cause, P2, and E-group independent review rounds addressed the interpretation, not a new measurement.

3 Early agreement and the strong-wind deficit

Regional statistics near domain-three activation were close, but a difference already appeared in the first tslist sample. The 06:11 gravity-wave-breaking-like downdraft can be compared qualitatively with the paper's mechanism (PDF pp. 6–8), without claiming a measured propagation velocity or unique phase pairing. The full-window exp-A point maximum of 46.06 m/s approaches the authors' 46.46 m/s, but the maxima belong to different pulses and locations. Similar model-point peak times are not station validation. The exp-A rendering of Fig. 5 accumulates from 05:15, whereas

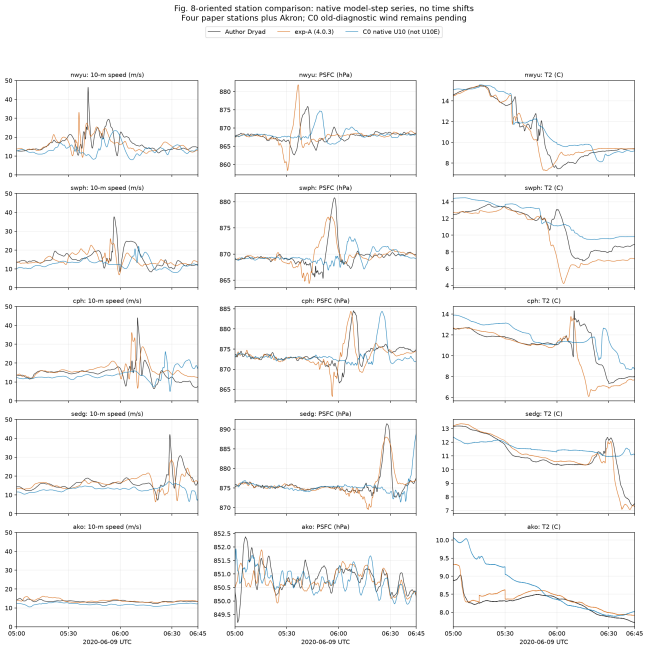


Figure 1: Four Dryad model tslist points and an Akron station; the four model points must not be interpreted as observations. Original: WRF [path]

Table 1: Cumulative event-region footprint of minute-resolved 10-m time-step maximum wind (05:15–06:30). Values are generated from the frozen JSON.

Threshold	Authors (km^2)	exp-A (km^2)	Ratio
$\geq$ 30 m/s	1036.8	810.6	0.782
$\geq$ 35 m/s	474.2	230.9	0.487
$\geq$ 40 m/s	96.1	18.3	0.19

the article starts at 03:30 (PDF p. 5); low-threshold areas across these figures are not directly comparable. The downdraft/thermal diagnostic is a same-UTC-time comparison with unavailable phase matching, not an intensity agreement.

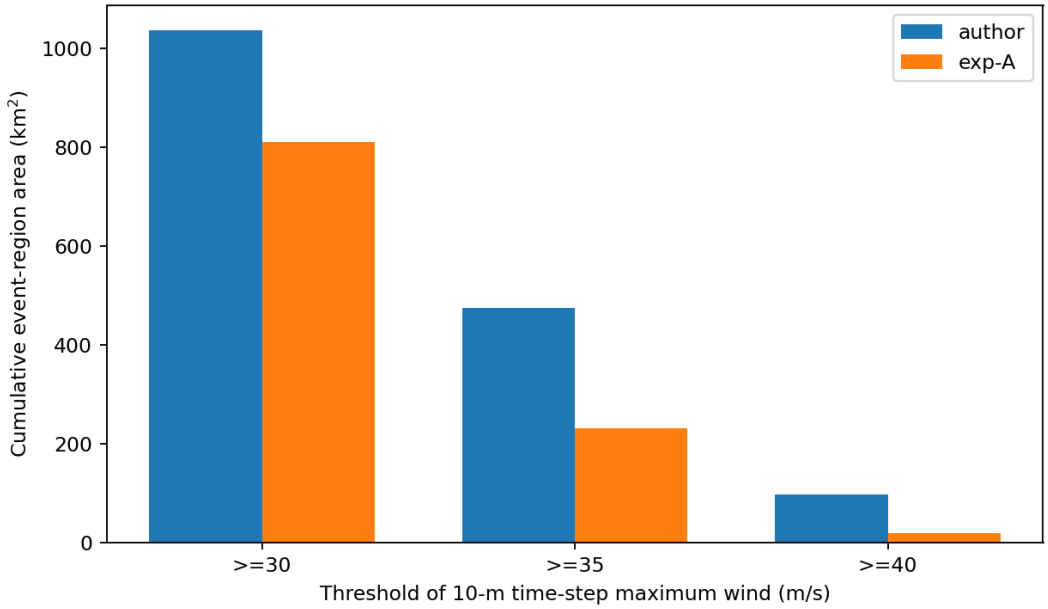


Figure 2: Cumulative, rather than instantaneous, strong-wind area versus wind threshold. Input: R16.4a exp-A numbers_20261020.json, accum.

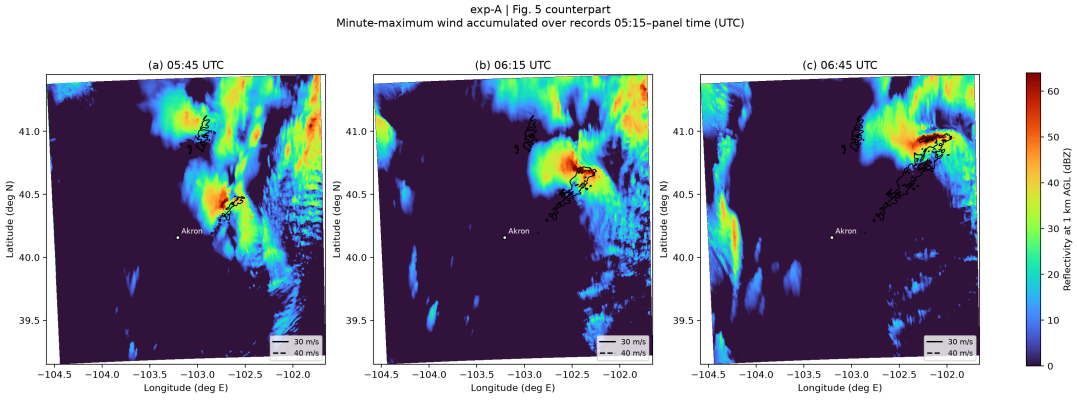


Figure 3: Exp-A rendition of Fig. 5, accumulated from 05:15; the article's Fig. 5 begins at 03:30. Original: WRF 06_Outputs.

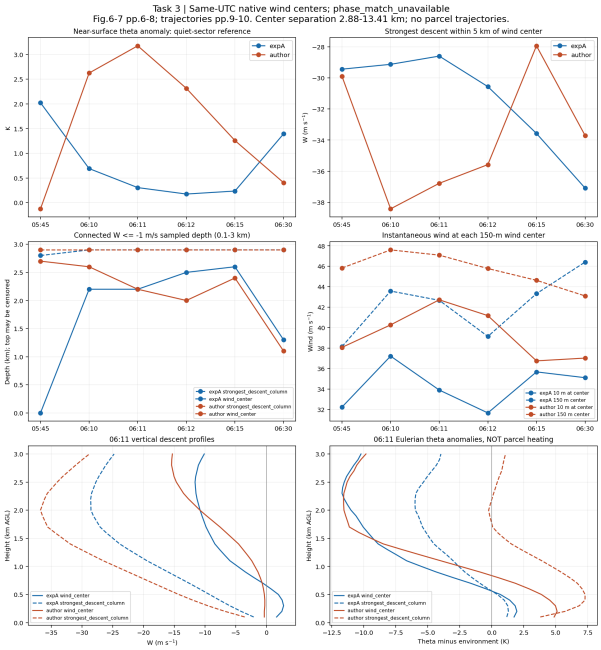


Figure 4: Descent and thermal diagnostic; visual mechanism resemblance does not close quantitative thermal or descent differences. Same-UTC comparison: phase_match_unavailable.

4 Perturbation members and vertical structure

Table 2: Frozen operator inputs: cumulative A, first-pulse B, and instantaneous lowest-layer area. Authors' B is a regional lower bound; P2's missing value is not a negative test.

Member	A (km^2)	B (m/s)	05:45 (km^2)
Authors	474.2	$\geq$ 46.46	9.39
exp-A	230.86	40.64	0.5
P2	181.31	39.67	missing
E2c	234.42	41.79	4.27
E2p1	44.86	41.46	0.0
E2p2	211.66	39.7	4.12
E2p3	107.72	41.8	0.0
E1	52.69	38.54	0.36

Every tested trajectory has B below the authors' lower bound, with a gap of at least 4.66–7.92 m/s across seven trajectories. The six-member B mean and standard deviation are 40.84 and 0.99 m/s; their standardized lower-bound separation is 5.7, a descriptive number rather than a probability because the members are few and non-independent. The authors' A is 2.02 times the largest tested A. Area varies widely across perturbations, and neither the area ratio nor this sample identifies a physical root cause. The largest lowest-layer area misses its coverage gate by 0.73 km^2; E2p1's B is 0.94 m/s below the weaker gate. P2 missingness limits the complete coverage assessment. The frozen rule labels the tested six-member family "systematically weaker", while E1 is assessed separately; a single E1 cannot exclude a boundary effect. At 05:45 the authors' 150-m maximum is 45.8 m/s versus the tested 33.9–42.9 m/s; at 06:10 the values are 47.6 versus 38.9–43.6 m/s. The respective authors' enhancements are 7.6 and 5.7 m/s, exceeding tested upper limits of 6.7 and 5.2 m/s. Of the preregistered readouts, 19 were comparable: 9 above, 8 within and 2 below the member range; 9 were unavailable. Other times need not follow the same ordering.

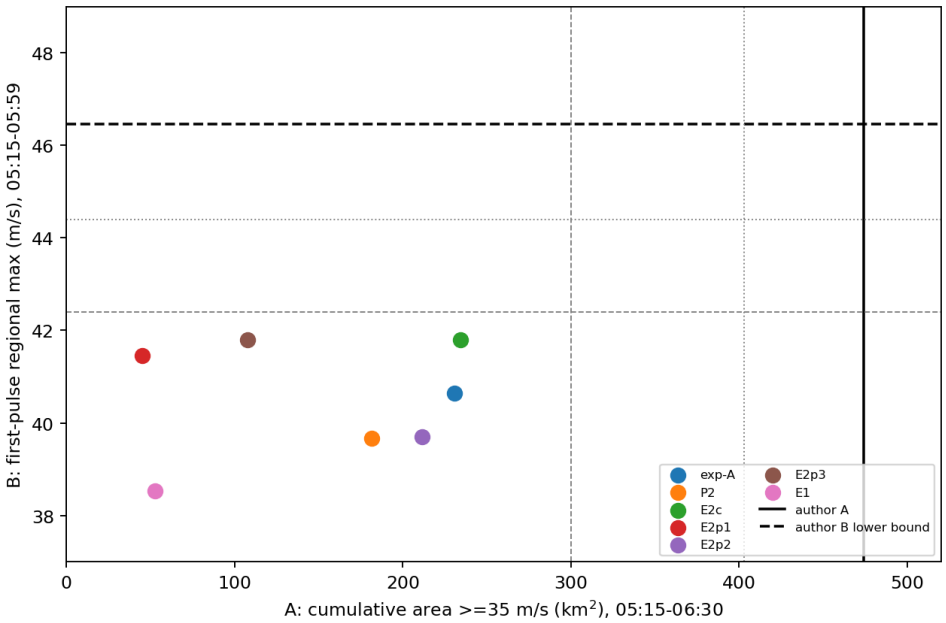


Figure 5: Member A–B scatterplot with the authors' A reference and B lower-bound reference; gray lines show frozen gates. A is cumulative area, B first-pulse regional wind. Input: R16.4b E1E2 numbers_20261020.json.

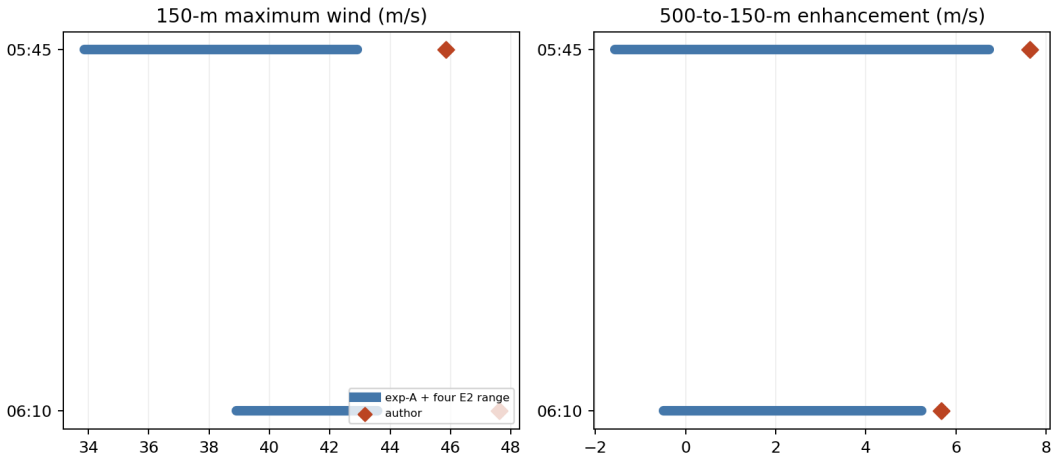


Figure 6: Preregistered 150-m wind and 500-to-150-m enhancement at 05:45 and 06:10. Bars span exp-A plus four E2 members; diamonds mark the authors. Same-UTC phase_match_unavailable. Input: readout_table.csv.

5 Observations, competing explanations, and adjudication

Observed Akron gust magnitudes vary in height and duration[1] (journal p. 2890, Table 2; physical PDF p. 16). The model quantity is a 10-m time-step maximum, so only magnitude can be discussed, not a co-located error. The nearest edge of the 35-m/s band to KAKO is 47.6 km for exp-A versus 51.2 km for the authors; for 40 m/s the distances are 79.2 versus 53.1 km. Both modeled swaths lie roughly 50 km northeast of the observed strong-wind region. Exp-A is weaker than the authors' simulation; compared with observations its regional peak is of similar order, but neither conservatism nor non-conservatism for observed extreme winds has been demonstrated. No universal amplification factor is warranted.

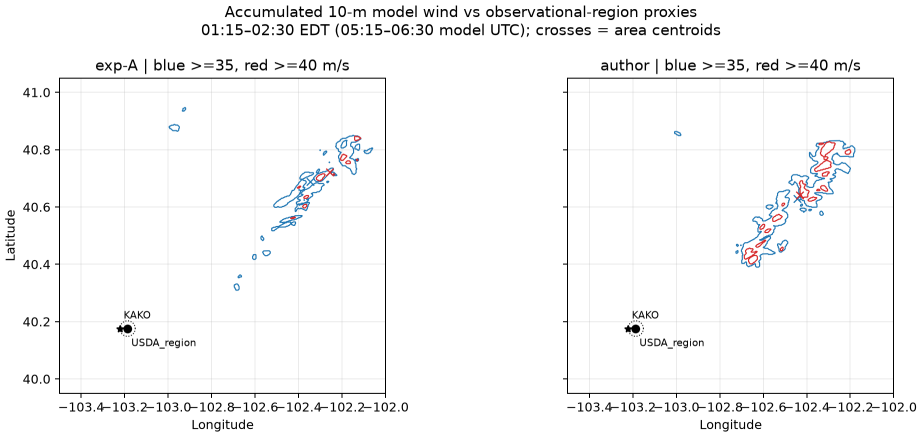


Figure 7: Location of modeled strong-wind swaths relative to Akron observations; edge distances are not co-located time-series errors. Original: expA_obs_comparison_20261010.

Table 3: Candidate explanations: a negative or unstable diagnostic does not exclude an entire mechanism.

Candidate	Current status
GFS cycle, namelist/eta, real version	Audited configuration only; authors' complete build remains unknown.
Thompson table, tslist joining, 06:00 restart	Specific suspicions examined, not a proof of numerical identity.
Translation and accumulation start	Registration improves overlap but not area ratios; low threshold depends on window.
Inversion and waveguide/critical layer	Diagnosis unstable after removing precipitation or changes sign.
Boundary version	One E1 combined boundary and start-chain change; no detected isolated effect, not excluded.
Precision and authors' build	Double-precision cold start failed; effect untested.

6 Failures, present scope, and restart entry

The bounded conclusion concerns this event, configuration, time windows and sampled perturbations: early regional statistics and some morphology are close; cumulative footprint and the first pulse are weaker; exp-A's later point maximum is not weak. The frozen "systematically weaker" label does not prove a cause. Measured propagation, phase match, and co-located observed time series are unavailable. E-member full-window point maxima have not been calculated, so exp-A's later maximum must not be generalized to the E family.

Table 4: Independent reviews and the superseded first-pulse interpretation.

Review	Adopted qualification and rejected inference
Root cause 86b59795	Six reviewers found no unique cause; repaired confounded author configuration, pressure provenance and accumulation window. MPI or a single boundary run was not accepted as the sole cause.
P2 b6ff70af / 8c51f4c6	Three preliminary and five formal reviewers: separate frozen label from causation; record cross-platform P2 and missing data. Rejected a claim that random divergence was excluded. An early reading attributed B to window truncation.
E group c5ffc06d	Six qualified supports. The authors' first-pulse value already reaches at least 46.46 m/s inside B's window; the authors' window-truncation reading was reversed and the difference is a lower bound. Member behavior at the window end remains unquantified; E1 does not eliminate boundary uncertainty.

Table 5: Operational failures and lessons, none a direct identification of the scientific cause.

Event	Disposition and lesson
SU under-reporting	SPR billing factor omitted, later corrected; verify platform rate.
P1/P2 core underuse; CRC startup failures	Separate memory-driven nodes from core use; test minimal communications and positive/negative boundary prechecks.
Sparse conservation copy	Minute frames later rescued to /work2 with sample hash comparison; preserve what diagnostics actually require.
Contaminated reviews; visual misreading	Invalid blind reviews excluded; quantitative readings returned to numerical source.
DP nested-init failure	Double input passed the former d02-open failure point, then wrf.exe segfaulted; source location unresolved and precision remains untested.

Restart entry

Table 6: Preserved entry points and open candidates. Costs below are historical estimates, not new compute requests.

Item	Location, evidence and known cost
Exp-A core	Keep TACC [path] `/expA_dryad_c0bdy_v1/`, approximately 425 GB. The original /scratch run root of approximately 422 GB has not been compared in full and must remain protected.
Frozen criteria	[path] `_ensemble.json` and [path] `reference/dp_member_20261010/frozen_criteria_dp.json`; Full SHA-256 values are recorded in the delivery audit; recheck original files before restart.
Double-precision build and inputs	S5 /scratch trial directories contain a successful build (approximately 0.89 GB) and converted input (approximately 17 GB) according to the inventory, not a current existence check. Locate the fourth wrf.exe crash. Debug build was estimated at 0.5 SU, short-test phase at about 25 SU. An entire DP member was estimated at 460–620 SU using an unmeasured 1.5–2 multiplier on the 307-SU single-precision cost; re-estimate from timed steps.
Boundary and dispersion	E1 changes boundary and cold-start chain together. New controlled members require a new design. Additional perturbations estimate spread but do not by themselves identify cause; historical CRC estimate is about 76 node-hours and 130 GB per member.
Other assets and authority	E members' original outputs were archived to PI desktop [path] P2 surface fields remain on CRC. S1 diagnostics of approximately 0.38 GB remain slated for preservation; the TACC /scratch inventory awaits PI disposal decision. This report moves and deletes nothing.

No curated card was found on how WRF compiler options or floating precision affect convection in this configuration; general predictability literature must not substitute for this missing evidence. Check source files, frozen gates, allocation balance and failure logs before designing a new experiment. Public attribution and storage disposition remain PI decisions.

Table 7: Historical compute use. TACC SUs and CRC node-hours are distinct units; estimates are not reconciled scheduler accounting.

Completed activity	Amount and source
TACC exp-A three segments	Approximately 307 SU on SPR, three nodes; main R16.4a report Section 8.
TACC cancelled P1/P2 chain	Approximately 93.2 SU; main report Section 8.
TACC postprocessing and DP build	Approximately 5 SU total, rough estimate.
CRC P2 B and C	Approximately 89 node-hours, derived in main report from four-node durations.
CRC E1/E2 chain	Approximately 410–460 node-hours, estimated, not individually billed.

Concluding remarks

The project remains paused, not closed, as of October 20, 2026. After new Claude and GPT models become available, return to frozen criteria and the double-precision crash site, then determine the next test and independent adjudication. `scientific_claim_approved=false`.

Numerical provenance and evidence status

All principal A, B, profile, KAKO-distance and ratio values are expanded by `numbers.tex`, generated directly from the two frozen `numbers_20261020.json` files. `NUMBER_LEDGER.tsv` maps each macro to its exact JSON key. Resource figures and operational incidents are separately attributed to the historical line reports; they are not presented as JSON measurements. The three new figures redraw PETW data, while historical PNGs are PETW internal assets, not publication-cleared figures. Evidence labels mean: directly read frozen JSON; partially checked line-report or decision-record statement without fresh remote raw-output verification; untested DP precision and co-located observational time series. PDF compilation is not scientific approval.

References

[1] S. J. Childs, R. S. Schumacher, and R. D. Adams-Selin. High-resolution observations of a destructive macroburst. *Monthly Weather Review*, 2021. DOI: https://doi.org/10.1175/MWR-D-20-0412.1; card GZYPXN4S, journal p. 2890, PDF p. 16.

[2] R. S. Schumacher, S. J. Childs, and R. D. Adams-Selin. Intense surface winds from gravity wave breaking in simulations of a destructive macroburst. *Monthly Weather Review*, 2023. DOI: https://doi.org/10.1175/MWR-D-22-0103.1; card 2DPBW6VX, PDF pp. 4–11.